



CSCI 430-01 Object-Oriented Development

Project 1

Stage 1 Group Report

Warehouse Object-Oriented System

Group 7

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Instructor: Prof. Sarnath Ramnath

September 30, 2026

1. Identification

Item	Value
Course	CSCI 430-01 Object-Oriented Development
Assignment	Project 1 - Stage 1 Group Report
System	Warehouse Object-Oriented System
Group	Group 7
Date	September 30, 2026
Instructor	Prof. Sarnath Ramnath
Private / group repository	https://github.com/km-fazle/Project1Stage1Group
Entry point	java UserInterface

2. Group Members

Group 7 completed Stage 1 of the warehouse project. The member list and contribution notes below are based on the work completed for the Stage 1 class design, individual class testing, and the WarehouseStage1 implementation.

Member	Role / contribution notes
K.M Fazle Rabbi	Stage 1 class design packet; Client and WishListItem individual work; Stage 1 WarehouseStage1 implementation and testing support. Repo creation, invited collaborators.
A S M Mukit	Group member for Stage 1 design review and group implementation coordination.
Pukar Puri	Group member; Stage 1 report & Design review group implementation.
Markose Mesay	Group member for Stage 1 design review and group implementation coordination.

3. Stage 1 Features Implemented

Stage 1 implements the following operations, plus the verification queries needed to show that the updates were stored correctly.

Stage 1 feature	Status
Add Client	Implemented
Add Product	Implemented
Add Product to Client Wishlist	Implemented
Display All Clients	Implemented
Display All Products	Implemented
Display Client Wishlist	Implemented

4. Implementation Summary

The Stage 1 implementation follows the Chapter 5 Library design approach. `UI` handles console input and output. `Warehouse` acts as the facade and controller for the business subsystem. `ClientList` and `ProductList` manage the two main collections. `Client` and `Product` represent the main domain objects. `WishlistItem` stores the product-quantity relationship for a client's wishlist.

The program is plain Java. It compiles with `javac *.java` and runs with `java UI`. New client IDs are generated as C1, C2, and so on. New product IDs are generated as P1, P2, and so on. A new client's account balance starts at 0.00. If a product is already on a client's wishlist, the old quantity is replaced; the quantities are not added together. Classes for later stages, such as `Order`, `Invoice`, and `WaitlistItem`, were not implemented.

Class	Responsibility
<code>UI</code>	Console menu, input, and output. Calls <code>Warehouse</code> only.
<code>Warehouse</code>	Facade/controller for Stage 1 business operations.
<code>ClientList</code>	Stores and searches <code>Client</code> objects.

Class	Responsibility
ProductList	Stores and searches Product objects.
Client	One warehouse client; owns that client's wishlist.
Product	One warehouse product with stock and sale price.
WishlistItem	Client-Product wishlist relationship and quantity.

5. Testing Summary

The group compiled the application and tested the required C1-C5, P1-P5, and wishlist scenarios from the Stage 1 handout. The application was also checked with Stage1RegressionTest. Selected results are listed below. The transcript is saved as stage1_script_session.txt in the group submission folder.

Required test	Result
javac *.java	Compiled with no errors
Create clients C1-C5	PASS
Display all clients with balance \$0.00	PASS
Create products P1-P5 with stock 10/20/30/40/50 and prices \$1-\$5	PASS
Display all products	PASS
C1 wishlist: P1, P3, P5 quantity 5	PASS
C2 wishlist: P1, P2, P4 quantity 7	PASS
C3 wishlist: P1, P2, P5 quantity 6	PASS
Add P3 and P5 quantity 7 to C2	PASS
C2 wishlist updated; C3 unchanged	PASS
java Stage1RegressionTest	ALL STAGE 1 REGRESSION TESTS PASSED

Commands used:

```
javac *.java  
java Stage1RegressionTest  
java UserInterface
```

6. Private Repository

Group repository URL:

<https://github.com/km-fazle/Project1Stage1Group>

The repository was forked from the professor's Project1Stage1Group starter. The Stage 1 code belongs in the WarehouseStage1 folder and must run with java UserInterface. The instructor (rsarnath) and TA (SajalCodeHub) has been added as collaborators with all the group members.

7. Testing / Implementation Issues

No blocking Stage 1 functional issues were found during the required test sequence. The wishlist quantity-replacement rule was verified: adding the same product again replaces the old quantity instead of adding the two quantities. Invalid client and invalid product cases fail cleanly. One repository follow-up remains: confirm that WarehouseStage1 is present on the group GitHub branch used for submission, that the repository is private as required by the handout, and that all group members, the instructor, and the TA have access.

8. Individual Work-Sharing Comments

The Stage 1 class design was prepared and submitted for the September 23 deadline. Individual class testing was completed for Client and WishlistItem using a separate driver, following the professor's BookTest example. The WarehouseStage1 Stage 1 program was implemented and verified against the required C1-C5 / P1-P5 wishlist sequence. Group members reviewed the Stage 1 design materials during group meetings. Specific work-sharing percentages are not listed here because the group did not adopt a formal percentage split for Stage 1.

9. Work-Sharing Percentages

Not required for this Stage 1 submission. No unequal work-sharing adjustment was requested by the group.

10. How to Compile and Run

1. Open a terminal in the WarehouseStage1 folder.
2. Run `javac *.java`.
3. Run `java UserInterface` for the interactive Stage 1 program.
4. Optionally run `java Stage1RegressionTest` for the automated Stage 1 checks.